

DCCN Lab Assignment - 6

Siddharth Narayan Mishra

123CS0189

DropTail

DropTail AQM Simulation

set ns [new Simulator]

set tf [open out.tr w]

\$ns trace-all \$tf

set qf [open queue_droptail.tr w]

Create nodes

set S0 [\$ns node]

set S1 [\$ns node]

set S2 [\$ns node]

set S3 [\$ns node]

set R1 [\$ns node]

set R2 [\$ns node]

set D0 [\$ns node]

set D1 [\$ns node]

set D2 [\$ns node]

set D3 [\$ns node]

Links

\$ns duplex-link \$S0 \$R1 10Mb 10ms DropTail

\$ns duplex-link \$S1 \$R1 10Mb 10ms DropTail

\$ns duplex-link \$S2 \$R1 10Mb 10ms DropTail

\$ns duplex-link \$S3 \$R1 10Mb 10ms DropTail

Bottleneck

\$ns duplex-link \$R1 \$R2 1Mb 20ms DropTail

\$ns queue-limit \$R1 \$R2 50

\$ns duplex-link \$R2 \$D0 10Mb 10ms DropTail

\$ns duplex-link \$R2 \$D1 10Mb 10ms DropTail

```
$ns duplex-link $R2 $D2 10Mb 10ms DropTail
$ns duplex-link $R2 $D3 10Mb 10ms DropTail
```

```
# Queue monitoring
set qmon [$ns monitor-queue $R1 $R2 $qf]
```

```
# TCP Connections
for {set i 0} {$i < 4} {incr i} {
    set tcp($i) [new Agent/TCP]
    set sink($i) [new Agent/TCPSink]
}
```

```
$ns attach-agent $S0 $tcp(0)
$ns attach-agent $D0 $sink(0)
$ns connect $tcp(0) $sink(0)
```

```
$ns attach-agent $S1 $tcp(1)
$ns attach-agent $D1 $sink(1)
$ns connect $tcp(1) $sink(1)
```

```
$ns attach-agent $S2 $tcp(2)
$ns attach-agent $D2 $sink(2)
$ns connect $tcp(2) $sink(2)
```

```
$ns attach-agent $S3 $tcp(3)
$ns attach-agent $D3 $sink(3)
$ns connect $tcp(3) $sink(3)
```

```
# FTP Applications
for {set i 0} {$i < 4} {incr i} {
    set ftp($i) [new Application/FTP]
    $ftp($i) attach-agent $tcp($i)
    $ns at 1.0 "$ftp($i) start"
}
```

```
$ns at 10.0 "finish"
```

```
proc finish {} {
    global ns tf qf
    $ns flush-trace
    close $tf
    close $qf
    exec nam out.nam &
    exit 0
}
```

```
}
```

```
$ns run
```

RED

```
set ns [new Simulator]
```

```
# Trace file
```

```
set tracefile [open out_red.tr w]
```

```
$ns trace-all $tracefile
```

```
# Queue output file
```

```
set qfile [open queue_red.tr w]
```

```
# Create Nodes
```

```
set S0 [$ns node]
```

```
set S1 [$ns node]
```

```
set S2 [$ns node]
```

```
set S3 [$ns node]
```

```
set R1 [$ns node]
```

```
set R2 [$ns node]
```

```
set D0 [$ns node]
```

```
set D1 [$ns node]
```

```
set D2 [$ns node]
```

```
set D3 [$ns node]
```

```
# Access links
```

```
$ns duplex-link $S0 $R1 100Mb 25ms DropTail
```

```
$ns duplex-link $S1 $R1 100Mb 25ms DropTail
```

```
$ns duplex-link $S2 $R1 100Mb 25ms DropTail
```

```
$ns duplex-link $S3 $R1 100Mb 25ms DropTail
```

```
# Bottleneck
```

```
$ns duplex-link $R1 $R2 10Mb 100ms RED
```

```
$ns queue-limit $R1 $R2 1000
```

```
# Output links
```

```
$ns duplex-link $R2 $D0 100Mb 25ms DropTail
```

```
$ns duplex-link $R2 $D1 100Mb 25ms DropTail
```

```
$ns duplex-link $R2 $D2 100Mb 25ms DropTail
$ns duplex-link $R2 $D3 100Mb 25ms DropTail
```

```
# RED parameters
Queue/RED set minthresh_ 50
Queue/RED set maxthresh_ 150
Queue/RED set q_weight_ 0.002
Queue/RED set linterm_ 10
Queue/RED set gentle_ true
Queue/RED set max_p_ 0.1
```

```
set link [$ns link $R1 $R2]
set queue [$link queue]
```

```
proc record {} {
    global ns queue qfile
    set time [$ns now]
    set qsize [$queue set curq_]
    puts $qfile "$time $qsize"
    $ns at [expr $time + 0.01] "record"
}
```

```
$ns at 0.0 "record"
```

```
for {set i 0} {$i < 4} {incr i} {
    set tcp($i) [new Agent/TCP]
    set sink($i) [new Agent/TCPSink]
}
```

```
$ns attach-agent $S0 $tcp(0)
$ns attach-agent $D0 $sink(0)
$ns connect $tcp(0) $sink(0)
```

```
$ns attach-agent $S1 $tcp(1)
$ns attach-agent $D1 $sink(1)
$ns connect $tcp(1) $sink(1)
```

```
$ns attach-agent $S2 $tcp(2)
$ns attach-agent $D2 $sink(2)
$ns connect $tcp(2) $sink(2)
```

```
$ns attach-agent $S3 $tcp(3)
$ns attach-agent $D3 $sink(3)
```

```

$ns connect $tcp(3) $sink(3)

# FTP Applications
for {set i 0} {$i < 4} {incr i} {
    set ftp($i) [new Application/FTP]
    $ftp($i) attach-agent $tcp($i)
    $ns at 1.0 "$ftp($i) start"
}

# Stop simulation
$ns at 10.0 "finish"

proc finish {} {
    global ns tracefile qfile
    $ns flush-trace
    close $tracefile
    close $qfile
    exit 0
}

$ns run

```

Adaptive RED

```

set ns [new Simulator]

set tf [open out_ared.tr w]
$ns trace-all $tf

# Nodes
set S0 [$ns node]
set S1 [$ns node]
set S2 [$ns node]
set S3 [$ns node]

set R1 [$ns node]
set R2 [$ns node]

set D0 [$ns node]
set D1 [$ns node]
set D2 [$ns node]

```

```
set D3 [$ns node]
```

```
# Access links
```

```
$ns duplex-link $S0 $R1 100Mb 25ms DropTail
```

```
$ns duplex-link $S1 $R1 100Mb 25ms DropTail
```

```
$ns duplex-link $S2 $R1 100Mb 25ms DropTail
```

```
$ns duplex-link $S3 $R1 100Mb 25ms DropTail
```

```
# Bottleneck
```

```
$ns duplex-link $R1 $R2 10Mb 100ms RED
```

```
$ns queue-limit $R1 $R2 1000
```

```
# Output links
```

```
$ns duplex-link $R2 $D0 100Mb 25ms DropTail
```

```
$ns duplex-link $R2 $D1 100Mb 25ms DropTail
```

```
$ns duplex-link $R2 $D2 100Mb 25ms DropTail
```

```
$ns duplex-link $R2 $D3 100Mb 25ms DropTail
```

```
# Adaptive RED Parameters
```

```
Queue/RED set minthresh_ 50
```

```
Queue/RED set maxthresh_ 150
```

```
Queue/RED set gentle_ true
```

```
Queue/RED set adaptive_ true
```

```
Queue/RED set cautious_ 1
```

```
Queue/RED set max_p_ 0.1
```

```
Queue/RED set targetdelay_ 0.02
```

```
# Get queue object directly
```

```
set link [$ns link $R1 $R2]
```

```
set queue [$link queue]
```

```
set qfile [open queue_ared.tr w]
```

```
# Record procedure
```

```
proc record {} {
```

```
    global ns queue qfile
```

```
    set time [$ns now]
```

```
    set qsize [$queue set curq_]
```

```
    puts $qfile "$time $qsize"
```

```
    $ns at [expr $time + 0.01] "record"
```

```
}
```

```
$ns at 0.0 "record"
```

```

# TCP Connections
for {set i 0} {$i < 4} {incr i} {
    set tcp($i) [new Agent/TCP]
    set sink($i) [new Agent/TCPSink]
}

$ns attach-agent $S0 $tcp(0)
$ns attach-agent $D0 $sink(0)
$ns connect $tcp(0) $sink(0)

$ns attach-agent $S1 $tcp(1)
$ns attach-agent $D1 $sink(1)
$ns connect $tcp(1) $sink(1)

$ns attach-agent $S2 $tcp(2)
$ns attach-agent $D2 $sink(2)
$ns connect $tcp(2) $sink(2)

$ns attach-agent $S3 $tcp(3)
$ns attach-agent $D3 $sink(3)
$ns connect $tcp(3) $sink(3)

# FTP
for {set i 0} {$i < 4} {incr i} {
    set ftp($i) [new Application/FTP]
    $ftp($i) attach-agent $tcp($i)
    $ns at 1.0 "$ftp($i) start"
}

$ns at 10.0 "finish"

proc finish {} {
    global ns tf qfile
    $ns flush-trace
    close $tf
    close $qfile
    exit 0
}

$ns run

```

